

Table 1001 to 1500 (continued)

The following pages reproduce the substance of the printed table 1001–1500 of the Pellian Equation report. For each entry the continued-fraction expansion of $\sqrt{a}$ is given on the upper line of the cell, the period being indicated by enclosing the central term (or pair of terms, marked with *) in parentheses; the lower line gives the second-row auxiliaries. The right-hand column gives the corresponding integer solution (y, x) of $y^2 - ax^2 = \pm 1$. Stars on the item-number mark entries where the period has a double middle term.

TABLE 1001 TO 1500—CONTINUED.

No.	Continued-fraction lines	$y \ / \ x$
1147	33; 8, 9, 34, 33, 1, 28, 27, 33, 26, 41, 3, (62) 58, 9, 34, 33, 1, 28, 27, 33, 26, 41, 3, ...	2789 45403 94471 524381986
1148	33; 1, 7, 2, (16) 59, 8, 31, (4)	[948]
1149	33; 1, 8, 1, 2, 3, (22) 60, 7, 44, 21, 20, (3)	1 65887 2 017226921547
1150	33; 1, 10, 3, 7, 4, 1, 2, (2) 61, 6, 21, 9, 14, 41, 21, (25)	349 25592 11843 84449
1151	33; 1, 12, 1, 1, 2, 2, 3, 6, 2, (33) 62, 5, 38, 29, 23, 25, 19, 10, 31, (2)	19426 07107 6 59056 71854707
1152	33; 63, 1, (1, 1) (a 4)	[unreadable]
1153*	33; 1, 21, 1, 1, 1, 6, 1, 7, 1, 1, 1, 1, 1, 2, 4, 1, 5, 2, 1, 3, (1, 1) 64, 3, 43, 24, 41, 9, 53, 8, 39, 27, 32, 29, 33, 24, 13, 48, 11, 23, 39, 16, (33, 33)	3 01789 02568 75073 102 47504 00230 72656
1154	33; 1, (32) 65, (2)	1155 341
1155	33; (1) (66)	34 1
1157*	34; (6, (81))	115374 17
1158	34; (3, (41))	4902 89385 75180 1 66914 55514 24551
1159	34; 22, 1, 2, 7, 4, 2, 2, 13, 4, 1, 3, 1, (2)	579
1160	—	—
1161	3, 45, 22, 9, 15, 25, 27, 5, 14, 45, 15, 42, (19)	56 21214 40972 1915 34168 54935
1162a	34; (17) (4)	2266 6136416723
1162b	34; 13, 1, 1, 1, 1, 2, 8, 7, 2, 4, 1, 3, (2) 5, 40, 27, 31, 32, 25, 8, 9, 29, 13, 45, 16, (27)	369 56541 12603 214090320
1162c	34; 11, 2, 1, 6, 1, (8) 6, 23, 42, 9, 54, (7)	1 68199 867 20773
1163	34; 9, 1, 2, 1, 2, 4, 1, 1, (33)	29599 61778
1164	7, 46, 19, 38, 23, 14, 31, 37, (2)	193 40870
1164b	34; 8, 1, 1, (16) 8, 33, 35, (4)	6604 27701 10943
1165*	34; 7, 1, 1, 3, 16, 1, 3, 1, (1, 1) 9, 36, 31, 19, 4, 51, 15, 36, (29, 29)	3 73828 22 6861725409
1166	34; 6, 1, 4, 2, 1, 1, 9, (6) 10, 49, 13, 25, 29, 38, 7, (11)	— —
1167	34; 6, 5, (11) 11, 13, (6)	— —
1168	34; 5, 1, 2, 7, (4) 12, 41, 23, 9, (16)	— —

TABLE 1001 TO 1500—CONTINUED.

No.	Continued-fraction lines	y / x
1169	34; 5, 4, 2, 3, 1, 1, 5, 1, 1, 1, 7, 1, (1) 13, [unreadable]	538 84918 28064 18423 59949 83913950
1170	34; 1, 29, 17, 32, 35, 1, 40, 25, 41, 8, 55, (7)	6499
1171	34; 5, 4, ((69))	—
1172	—	—
1173	34; 4, 49, 51, ... 15, —	15274 30263 5 22684
1174	34; — 13, 42, 21, 26, 17, 35, 30, 27, 41, (2)	1233 5769183900 20614397
1175	34; 16, 33, 34, 3, 1, 1, (16) 16, 31, 37, (4)	— —
1176	34; (4, 4) 1, 43, —	4
1177	34; 18, 15, 14, 1, 4, 2, 13, 3, 1, 22, 11, 2, 1, 1, 1, 6, 4, 2, 2, 1, 1, 7, (34)	1363 67209 12136 61406 74774
1178	3, 43, 1, 2, 6, 13, 30, 5, 17, 50, 3, 6, 25, 34, 27, 39, 10, 15, 26, 23, 30, 37, 9, (2)	46724 42879 20656 93035 91365
1179	34; 19, 3, 2, 5, (1) 34, 25, 11, (50)	12901 4 42212440
1180	34; 20, 25, 31, (24)	4801
1181*	34; 21, 16, (2) 22, 9, 1, 3, 7, 2, 1, 2, 1, 1, 2, 2, 2, 8, (6)	276 38354 60657 78460
1182	—	9482 01015 39044
1183	3, 31, 3, 7, 48, 17, 9, 24, 37, 21, 33, 32, 23, 24, 27, 8, (11)	—
1184	—	751375316
1185	34; 22, (2)	309
1186	34; 23, 9, (33) 24, 45, —	10610 20 24999
1187	—	58950
1188	—	—
1189*	— 1, 20, 1, 36, 31, 21, 39, (20)	— —
1189b	34; 25, 2, 1, 3, 3, 5, 1, 16, 2, 1, (13, 13) 37, 29, 1, 41, 17, 20, 11, 55, 4, 23, 44, (5, 5)	35 62788 38045 1224 37647 36718
1189c	34; 26, 29, 1, 2, 2, 1, 2, 1, (10) 34, 23, 22, 39, 19, 47, (6)	2727 5922568872 37
1189d	34; 27, 33, 2, 7, 22, 1, (3)	1305
1189e	38, 1, 9, (17)	96401 76328
1189f	34; 28, 25, 2, 16, 49, 7, (4) 9, 3, 53, (14)	9655220815 41142 0138725010
1189g	34; 2, 21, 1, 3, 1, 1, 2, 3, (4) 29, 3, 16, 35, 31, 24, 19, (15)	— 1 2803159225
1189h	34; 2, 17, 1, 1, 4, (34) 30, 1, 34, 15, (2)	63 43775 11347
1189i	34; 3, 13, 33, 4, 9, 1, 1, 1, 2, 1, (33) 31, 14, 7, 41, 26, 37, 19, 49, (2)	3 90951 47851564921960 —
1189j	34; 2, 32, 17, 3, 20, 2, 1, 2, 16, 1, 6, 1, 2, (1, 1) 33, 5, 3, 23, 36, 25, 4, 57, 9, 45, 20, (33, 33)	24 58219 58945 847 64031 17418

TABLE 1001 TO 1500—CONTINUED.

No.	Continued-fraction lines	y / x
1190	34; (4) (34)	2 69
1191	34; 1, 34, (3)	3175
1192	—	97167
1193*	34; 1, (22)	33 54723487 3792842959

No.	Continued-fraction lines	$y \ / \ x$
1194	—	—
1195	34; 35, 2, 1, 3, 2, 1, 1, 1, 1, 7, (17) 24, 39, 17, 24, 33, 31, 28, 39, 9, (4)	8549 5380952 92268
1196a	34; 36, 33, 1, 1, 1, 1, 3, 2, 8, (5, 5) 1, —	6264 2 16474 678674590
1196b	34; 37, 32, 1, 1, 4, 7, 16, 29, 8, (13, 13) 9, 1, 1, 1, 6, 3, 1, (10)	1345 1660293529 46500 415
1196c	34; 19, 5, 7, 42, 25, 41, 10, 17, 49, (6)	12 25
1196d	34; 38, 31, 2, 7, 2, 1, 1, 10, 1, (12)	—
1196e	34; 39, 29, 3, 9, 26, 29, 39, 6, 59, (5) 30, (2, (23))	— —
1197	40, 28, 27, 16, 1, (8)	8784589290
1198	—	5116549690
1199	34; 1, 4, 59, (7)	1 21525 49298 36463 39
1200	—	42 06256 96350 23592
1200b	41, 1, 27, 1, 2, 1, 3, 3, (34) 1, 34, 21, 38, 23, 18, 29, 6, 7, 11, 14, 3, 58, 9, 40, 19, 42, 17, 21, (2)	4917473 75599 77820
1200c	34; 42, 26, 25, 31, 2, 9, 1, 1, 13, (3)	1351
1200d	43, 35, 25, 7, 34, 35, 5, (22)	—
1200e	34; 44, 24, (16)	—
1200f	34; 1, 2, 23, 19, 9, 4, 4, 2, 1, 1, 1, 7, 13, 1, 2, 1, 2, 1, 1, 4, 22, 1, 7, 1, 7, 16, 15, 24, 35, 27, 40, 9, 5, 48, 19, 40, 21, 32, 35, 10, 3, 59, 8, 45	— —
1201*	34; 45, 2, 43, (2, (52, 252))	71 85416 85609 44230
1202	—	2490 13832 32746 50571 7474388352
1203	34; 1, 23, (34)	10817
1204	1, 2)	312
1205	—	—
1206	22, (2) 1, (33)	—
1207a	46, 1, 11, —	4 3142258212
1207b	34; 1, 2, 2, 57, (2)	16 95565 54116
1207c	34; 47, 21, 20, 7, 2, 2, 3, 4, 22, 1, (8) 29, 9, 27, 25, 19, 16, 3, 60, (7)	588 33925 37695 2 06668 74089
1207d	48, 20, 16, 1, (12)	1 454922052 33387
1207e	34; 1, 24, (63, 14) (5)	11
1207f	49, 19, 2, 3, 1, (33) 1, 25, (2)	— —
1207g	34; 50, 18, 38, 1, 9, 59, (2)	17611 5293902486
1208	51, 17, 47, 1, 2, (8)	42
1209	23, 44, 23, (8)	4706165263
1209b	34; 52, 16, 1, 3, 2, 1, 1, (22) 53, 3, 9, 40, 17, 25, 29, 40, (3) 7, —	29335 570 —

TABLE 1001 TO 1500—CONTINUED.

No.	Continued-fraction lines	$y \ / \ x$
1210	34; 1, 3, 1, 1, 1, 7, 1, 1, (6) 54, 15, 39, 26, 41, 9, 6, 31, (10)	430 99524 14992 19281
1211	34; 1, (3) 55, (14)	5 174 3 65980

No.	Continued-fraction lines	y / x
		127 41151
1212	34; 1, 4, 2, 1, 2, 2, 1, (16) 56, 13, 24, 37, 23, 21, 47, (4)	20 27117 706 00734
1213*	34; 57, 12, 49, 13, 33, 36, 7, (27, 27)	7321059 0947790823 6704190745 7
1214	34; 1, 5, 2, 1, 6, 3, 1, 1, 13, 2, 1, 2, 2, (34)	244
1215	58, 11, 23, 43, 10, 19, 31, 38, 5, 25, 37, 22, 29, (2)	1916 03685
1216	—	66814 48801
1216b	34; 1, (5) 59, (10)	7 91969 276 28235469
1216c	34; 1, 6, 1, 3, 4, 2, 1, 1, 4, (17) 60, 9, 48, 17, 15, 25, 31, 36, 15, (4)	10 —
1217*	34; 1, 7, 1, 2, 1, 3, 1, 1, 1, (1, 1)	—
1218	61, 8, 47, 19, 43, 16, 37, 29, 32, (31, 31)	—
1218b	34; 1, (8) 62, (7)	— —
1219	1, 10, 1, 1, 1, 7, 9, 1, 5, 2, 4, 5, 6, 1, 3, (1)	26 76791 47002 595513423815694
1220	34; 63, 6, 43, 25, 42, 9, 7, 54, 11, 29, 15, 13, 10, 49, 15, (46)	81111 24548
1221	34; 64, (5) 1, 16, (2) 65, 4, (33)	— 21368152 22 12495 24122 773 42454 09307
1222	34; 1, 22, 3, 7, 2, 3, 1, 1, 1, 4, 2, 1, (4)	122345
1223	66, 3, 22, 9, 29, 17, 38, 27, 39, 14, 23, 42, (13)	35 1
1223b	34; 1, (33) 67, (4)	35 1
1224	34; (1) (68)	122365 44730 59716 99506 15 67486 75428 71047
1226*	35; (7, (01))	21409103 14
1227	35; (35)	—
1228	35; 23, 2, 1, 7, 8, 1, 1, 1, 2, 2, 1, 1, 5, 3, 1, (16)	491
	3, 24, 43, 9, 8, 41, 27, 36, 23, 24, 31, 37, 12, 17, 51, (4)	584 79034 52350 15227 51177 20517 72574 24010 98134 87200
1229*	35; 17, (1, 1)	—
1230	4, (35, 35)	—
1230b	35; (14) (5)	—
1231	35; 11, 1, 2, 7, 2, 4, 1, 13, 4, 1, 1, 1, 1, 6, 2, 2, 4, 3, 1, 2, 23, (35) 6, 45, 23, 9, 30, 13, 54, 5, 25, 37, 30, 29, 39, 10, 27, 26, 15, 18, 39, 25, 3, (2)	— —

TABLE 1001 TO 1500—CONTINUED.

No.	Continued-fraction lines	y / x
1232	35; (1, (07))	210 76139 064889938541
1233	8, 49, 17, 16, 9, 37, 32, 19, 11, 7, 56, (9)	7400 75542 46657
1234	—	1669 10230 73856
1235	35; 7, 1, 3, 1, 4, 4, 2, 9, 1, 1, 2, 3, 1, 1, (34) 9, 50, 15, 47, 14, 15, 30, 7, 39, 30, 25, 18, 31, 39, (2)	58632 78690 67265 —
1235b	35; (7) (10)	7 246 18261 82722 6 42027 25495

No.	Continued-fraction lines	y / x
1236	35; 6, 2, 1, 1, 1, 4, 1, 3, 1, 1, 2, 1, (22)	30 80599 95913 30265
	11, 25, 35, 28, 39, 13, 47, 16, 35, 33, 20, 49, (3)	1083 47814 40276 61982
1237*	35; 5, 1, 5, 1, 1, 3, 1, 1, 2, 23, 17, 1, 1, 5, 2, 1, (7, 7)	25650
1238	12, 51, 11, 36, 33, 17, 36, 31, 27, 3, 4, 37, 33, 12, 23, 44, (9, 9)	9 02501
1239	35; 5, 2, 1, 1, (34)	5
	13, 26, 29, 41, (2)	176
1239b	35; (5) (14)	24102
		8 48719
1240	35; 4, 1, 2, 7, (2)	9 65005
	15, 41, 24, 9, (31)	339 95032
1241*	35; 4, 2, 1, 1, 3, 8, (1, 1)	11780
1242	16, 25, 32, 35, 19, 8, (35, 35)	4 15151
1242b	35; 4, 7, 1, 1, (2)	4875
	17, 9, 38, 31, (23)	1 71874
1243	35; 3, 1, 9, (3)	1616 45760 60360
	18, 49, 7, (22)	57013 08077 08799
1244	35; 19, 40, 23, 20, 11, 8, 43, 25, 44, 5, 31, 13, 55, (4)	2 13528
		75 34241
1245	35; 3, 1, 1, 17, (14)	1257 54312 87288
1246	20, 31, 39, 4, (5)	44389 66931 48735
1246b	35; 21, 22, 45, 9, 10, 57, 5, 18, 47, 11, 42, 25, 45, (2)	26313
		9 29118589
1247	35; 3, 5, 9, (1)	5617
1248	22, 13, 7, (58)	2 61326 40028 30963 92593
		92 35587 03432 90296 88240
1248b	35; 23, (4)	2 757880071
		21 92943
1249*	35; 2, 1, 13, 2, 7, 2, 1, 2, 3, 1, 3, 1, 1, 1, 4, 1, 3, 1, 8, 23, 2, (4, 4)	775 63250
1250*	24, 45, 5, 32, 9, 25, 37, 24, 17, 45, 16, 39, 27, 40, 13, 48, 15, 57, 8, 3, 31, (15, 15)	90969 00792 82080
1251	—	32 18812 08291 34849
1251b	35; 2, 1, 4, 2, 1, (1, 1)	—
	25, 41, 14, 25, 34, (31, 31)	—
1251c	35; 2, 1, 2, 2, 2, 5, (35)	—
	26, 37, 23, 25, 27, 13, (2)	—
1252	27, 33, 32, 29, 39, 12, 19, 33, 36, 13, 52, 9, 3, 32, 11, 57, (4)	—

TABLE 1001 TO 1500—CONTINUED.

No.	Continued-fraction lines	y / x
1253	35; 1, 1, 1, 17, (10) 28, 29, 41, 4, (7)	78320
		27 72351
		10234
		3 62405
1254	35; 1, 2, 2, 1, (34)	87 88466 34708
	29, 25, 11, 49, (2)	3113 40025 72889
1255	35; 2, 2, 1, 7, 6, 3, 4, 1, 2, 1, 11, (14)	11075
	30, 21, 46, 9, 11, 21, 14, 45, 19, 49, 6, (5)	3 92499
1256	35; 2, 3, 1, 2, (17)	2860
	31, 17, 40, 25, (4)	1 01399
1257	35; 2, 4, 1, (22) 32, 13, 56, (3)	3233
		1 14669
		313 98432 81069
		11140 91438 32810
1258*	35; 2, 7, 2, (1, 1)	2

No.	Continued-fraction lines	y / x
	33, 9, 26, (33, 33)	694 50461 60725 24662 24909 57682
1259	35; 1, 13, 1, 2, 3, 2, 1, 1, 5, 1, 6, 4, (35) 34, 5, 47, 22, 19, 25, 31, 38, 11, 53, 10, 17, (2)	17412 0470781 6 18556 69263
1260	35; (2) (35)	4004 1 42297
1261*	35; 1, 1, 23, 5, 1, 7, 17, 1, 1, 1, 2, 5, (1, 1) 36, 35, 3, 14, 53, 9, 4, 43, 27, 36, 25, 12, (35, 35)	360 12799
1262	37, 34, 7, 43, 26, 41, 13, 49, 14, 17, 53, (2)	2 065989290
1263	35; 1, 1, 5, 1, (22) 38, 33, 1, 58, (3)	149 86708 5332 40465
1264	35; 1, 1, 4, (4)	175 73127
1265	39, 32, 15, (16)	6255 14462
1265b	35; 1, 1, 3, 4, (6) 40, 31, 19, 16, (11)	69835 99490 24 86789 07849
1266	35; 1, 1, 2, 1, 1, 2, 3, 1, 3, 1, (34)	34 299621654
1267	41, 30, 23, 34, 33, 25, 17, 46, 15, 55, (2)	1 63267 43118 58 18371 13691
1267b	35; 1, 1, 2, 7, 1, 1, 23, (5) 42, 29, 27, 9, 34, 37, 3, (11)	32799 912007
1268	35; 1, 1, 1, 1, 3, 1, 5, 1, 2, 4, 9, 1, (16) 43, 28, 31, 37, 16, 49, 11, 44, 23, 16, 7, 61, (4)	3 —
1269	35; 1, 1, 1, 1, 1, 7, 3, (2)	5 14164 62158 51476
1270	44, 27, 35, 28, 41, 9, 20, (27)	183 44944 08681 34807
1270b	35; 1, 1, 1, 3, 1, 1, 1, 1, 5, 1, 7, (14) 45, 26, 39, 69, 6, 41, 29, 30, 39, 11, 54, 9, (5)	— —
1271	35; 1, 1, 1, 6, (2) 46, 25, 43, 10, (31)	— —
1272	35; 1, (1)	—
1273	47, (24)	—
1273b	35; 1, 2, 8, 1, 1, 2, 2, 4, 23, 1, 1, 3, 1, 2, 5, 7, 1, (2) 48, 23, 8, 39, 31, 24, 27, 16, 3, 39, 32, 17, 41, 24, 13, 9, 48, (19)	— —

TABLE 1001 TO 1500—CONTINUED.

No.	Continued-fraction lines	y / x
1274	35; 1, 1, 1, 6, 2, 1, 2, (2)	1571240
1275	49, 21, 49, 10, 25, 38, 23, (26)	[59441 8 004890919
1276	—	02580
1277*	35; 1, 3, (2, (52))	3329114 9178452473 36799
1278	50, 20, —	931 440647297 42285 48482
1279	—	—
1280	35; 1, 1, 2, 2, 2, 7, 1, 1, 9, 1, 2, 13, 1, (16)	—
1281	51, —	—
1282	33, 24, 25, 28, 9, 35, 36, 7, 45, 24, 5, 63, (4)	—
1283	—	—
1284	35; 3, 2, 5, 17, 1, 2, (6, 6) 17, 29, 13, 4, 47, 23, (11, 11)	— —
1285*	—	4
1286	—	143
1287	52, 44, (2)	—
1288	—	—
1289*	53, (18)	—
1290	35; 17, 15, 1, 1, 13, 1, 3, 23, 1, 1, 2, 2, 1, 5, 1, 3, 1, 11, 7, 3, 44	7 89404 78828 27783 39935 75041

No.	Continued-fraction lines	y / x
1291	1, 33, 38, 19, 15, 51, 18, 3, 41, 30, 25, 22, 45, 11, 50, 15, 53, 6, 9, 55, 16, 37, 1,	282 31569 87094 18598 91612 561040409
1292	54, 1, 1, 1, 2, (35) 1, 1, 30, 33, 31, 34, 27, (2)	— —
1292b	35; (1, (74)) 55, —	220 51841 81748
1292c	35; 3, 1, 2, 13, 1, (22) 1, 3, 1, 16, 41, 25, 5, 64, (3)	29175 —
1292d	56, 47, 1, 3, 9, 1, (34) 3, 9, 49, 18, 7, 63, (2)	7903 2467410338 34103
1292e	35; 57, 13, 1, 4, 1, 9, 1, 2, 4, 1, 3, 2, 2, (35) 1, 37, 7, 46, 23, 14, 47, 17, 26, 29, (2)	5 15591522 61505 65140 88226
1292f	35; 58, 4, — 65, —	184 24
1292g	19, 1, 1, 6, 24, 34, — 11, —	2156 877
1292h	35; 1, 7, 2, 2, 1, 17, 4, 1, 2, 1, (1, 1) 10, 3, 53, 36, 9, 29, 21, 49, 4, 15, 44, 21, 36, (31, 31)	48426 68507 94225 5674372178887
1292i	35; 5, 61, 2, 1, 2, 13, 1, (34) 62, 25, 37, 26, 5, 65, (2)	2 53631 10565 7072 8
1292j	35; (9) 5, 63, ((76))	9 323
1292k	35; (8) 40, 19, 2, 8, 1, 1, 1, 1, 2, 1, 1, 13, 1, 3, (1, 1) 64, 25, 8, 41, 29, 32, 35, 23, 31, 40, 5, 53, 16, (35, 35)	835 94139 2313980530 30012 50804
1292l	35; 65, 7, 1 (6)	12 431
1292m	35; 66, (10) 1, 1, 2, 3, 1, 1, 1, 1, 4, 1, 11, 6, 2, 4, 3, 23, 1, 1, 15, 33, 34, 25, 18, 37, 31, 30, 39, 13, 55, 6, 11, 30, 15, 22, 3, 45, 26, 1, 7, 1, 3, 2, 1, 6, 2, (35) 9, 50, 17, 23, 45, 10, 33, (2)	94899 51267 74456 1974 27745 80291 79860 70936 75339 57910 18687 08410
1292n	35; (16) 4, (4)	64178 41

TABLE 1001 TO 1500—CONTINUED.

No.	Continued-fraction lines	y / x
1293	35; 1, (22) 68, (3)	24 863
1294	35; 1, (34) 69, (2)	1295 36
1295	35; (1) (70)	1 36
1297*	36; (7, (20))	129337661 24
1298	36; (36) (24)	—
1299	36; (3)	86158 649
1300	36; (18) (4)	—
1301*	36; 14, 2, (2, 2) 5, 29, (25, 25)	6025 2 17318
1302	36; (12) (6)	43132
1303	36; 10, 3, 2, 1, 23, 2, 1, 2, 1, 3, 3, 1, 1, 7, 2, 5, 11, 1, 5, 1, 1, 1, 4,	10127 26314 51893 46887 71461

No.	Continued-fraction lines	y / x
	7, 21, 22, 49, 3, 26, 39, 21, 43, 18, 19, 33, 38, 9, 31, 13, 6, 57, 11, 42, 27, 41, 14, 1, 1, (38) 33, 39, (2)	3 65564 74232 60608 71132 28808 — —
1304	36; (9) (8)	2899 325
1305	36; (8) (9)	8
1306*	36; 7, 4, 1, 2, 11, 1, 2, 4, 2, 9, 1, (7, 7)	862 68151 05389
1307	10, 15, 42, 25, 6, 47, 23, 15, 31, 7, 58, (9, 9)	31176 12101 50515
1308	36; 11, 37, 34, 17, 43, 22, 31, 41, (2)	1957 48028127 6
1308b	36; (6) (12)	—
1309	13, 36, 35, 15, 19, 36, 33, 25, 21, 4, 55, 15, 52, 9, (17)	991 09921 7 99544 80317 01740 72265
1310	36; 5, 6, 2, 1, 1, 1, 1, 1, (6) 14, 11, 26, 35, 31, 34, 29, 41, (10)	34 56382 1251 00021
1311	36; 4, 1, 4, 2, (1)	17
1312	15, 49, 14, 25, (38)	5353 1 931842607
1312b	36; 4, 1, 1, (17) 16, 33, 39, (4)	53137 616
1313*	36; (4, 4) (17, 17)	58

TABLE 1001 TO 1500—CONTINUED.

No.	Continued-fraction lines	y / x
1314	(4) (18)	1454 924448
1315	36; 20, 46, 15, 6, 59, 9, (10) (2)	8875 7141365416
1316	1, 1, 1, 1, 2, 3,	—
1317	21, 37, 31, 52, 35, 25, 19, (28)	65508
1318	2, 3, 1, 5, 3, 1, 1, 1, 4, 1, 17, 3, 10, (24)	23
1319	3, 28, 17, 49, 12, 19, 39, 28, 41, 13, 57, 4, 23, 7, (3)	8716540105
1319b	22, 19, 31, 36, (5, 43) 17, 21, 14, 53, 9, 18, 51, 7, 42, 29, 33, 34, 27, 6, 49, 21, (2) 38, 11, (2) 23, 1, 6, 1, 13, 1, 1, 1, (35) 10, 59, 5, 46, 25, 47, (2)	1 03450 99306 37 54287 70770 93751 3138 36679 69954 08957 1 13936 11468 26276 81762 974655147 334 12130 01011 59214009
1320	36; (23, (43))	3
1321*	25, 1, 1, 2, 1, 8, 2, 2, 1, 1, 3, 1, 23, 2, 4, 2, 1, 4, 6, 2, 1, 1, 7, 2, (14, 14)	14 70631 38627 6169162171 69925
1322*	26, 45, 8, 29, 24, 33, 37, 16, 55, 3, 32, 15, 24, 43, 15, 11, 27, 31, 40, 9, 33, (5, 5)	534 50926 71539 96432
1323	—	29885211
1324	36; 1, 3, 1, (1, 1) 41, 17, 38, (31, 31)	— —
1325*	27, 1, 2, 7, 1, 2, (2)	1 00360
1326	37, 26, 9, 47, 22, (27)	50101
1327	—	—
1328	28, —	—
1329	—	—
1330	1, 1, 2, 2, 3, 4, 1, 1, 3, 1, 2, 1, 2, 5, 1, 2, 3, 8, 1, 3,	36 21799
1331	33, 35, 24, 27, 20, 15, 36, 35, 17, 44, 21, 40, 25, 12, 45, 23, 21, 8, 53, 15, 56, 15, 51902,	12718 55353 72450 38443 74810

No.	Continued-fraction lines	$y \ / \ x$
1332	2, 23, 2, 6, 7, 1, 13, 1, 2, 10, (18)	42650 10838 17312 41819 40987
	29, 3, 33, 11, 9, 60, 5, 47, 24, 7, (4)	—
1332b	36; (29) (2) 2, 25, (26)	—
		5
		182
1332c	36; 30, 2, 1, (2, (35)) 21, 51, (2)	2549
		178750
		65032649
1332d	3, 1, (3) 17, 47, (16)	13447
		—
1332e	36; 32, 5, 8, 1, 13, 1, 2, 4, 4, 1, 1, 1, 2, 2, 1, 1, 6, (24) 13, 8, 61, 5, 34, 18, 23, 16, 15, 40, 29, 37, 24, 25, 32, 39, 11, (3)	—
		19 32883 59509 0314419156
1332f	36; 33, 7, 1, 1, 1, 1, (4) 9, 41, 30, 31, 39, (14)	704 64145 63882
		344615829
		12
1332g	34, 14, 10, 2, 1, 4, 1, 1, 6, 1, 2, 1, (35) 5, 7, 25, 43, 14, 35, 37, 10, 49, 19, 53, (2) 35, — (36)	10727
		273 0230738710 19890
		—
		9961
		73

TABLE 1001 TO 1500—CONTINUED.

No.	Continued-fraction lines	$y \ / \ x$
1333	36; 1, 1, 1, 23, 1, 1, 5, 7, 1, 17, 2, 1, 1, 1, 5, (2)	904 20166 84140
1334	37, 36, 1, 59, 12, 9, 61, 4, 27, 36, 29, 41, 12, (31)	33012 64933 32151
1335	36; 13, 1, 2, 2, 1, 1, (2)	31585694 08256
1336	5, 49, 22, 25, 34, 35, (23)	721029355
1337	38, 35, 1, 79, (7)	—
1338	—	—
1339	—	—
1340	—	—
1341	39, 34, 15, (10)	45124
1342	19, 25, 39, 24, 23, 44, 15, 12, 55, 9, 20, 33, 39, (8)	174 56099
1343	—	6380 43737 9199662925
1344	40, 33, 2, 1, 8, 2, 4, (10)	373 31647
1345	11, —	13674 97272
1346	41, 32, 23, 31, 23, 2, 47, 8, 31, 16, (7)	4
1347	1, 2, 10, (12)	173
1348	—	73194
1349	42, 31, 27, 39, 26, 7, (6)	08813
1350	4, 1, (4)	4104
	14, 51, (13)	1 50175
1351	36; 43, 23, 19, 1, 11, 61, (18)	7 19932
	40, (4)	4259362310
1352	44, 28, 11, 31, 1, 1, 2, 3, 3, 1, 1, 3, 1, 2, 1, 7, 2, 2, 17, 1, 10, 1, 1, 14, (8)	36359 14354 86763
1353	35, 31, 36, 25, 20, 19, 35, 36, 17, 45, 20, 49, 9, 28, 29, 4, 63, 7, 36, 37, 5, (9)	13 31459 72360 20892 39201
	45, 27, —	109390
1353b	36; 46, 1, 26, 39, (22)	8180152981
	(2)	165
1353c	4, 1, (35)	3
1353d	47, 25, 43, 2, 13, 59, (2)	—
1353e	36; (17)	—
	48, 24, 47, (4)	6049

No.	Continued-fraction lines	y / x
	(14)	132 4841
1353f	36; 49, 23, 1, 24, 2, 1, 9, 1, 2, (36) 12, 1, 2, 55, 7, 46, 25, (2)	3405 82362 60 50115
1353g	36; 50, 22, 23, 4, 2, 1, 5, 1, (35) 46, 1, 11, 61, (2)	1 64753 92 8426367862 56 21145 67828 13793 2063 81035 31297
1353h	51, 21, 2, 20, 1, 23, 1, 9, 1, 1, 7, 1, 1, 1, 2, 1, 5, 2, 1, 1, 5, (18) 1, 41, 3, 64, 7, 37, 36, 9, 43, 28, 39, 21, 47, 12, 27, 32, 39, 13, (4)	5 55200 34840 203 91806 65951
1353i	36; 52, 32, —	12804 4 70414698
1353j	36; 2, 5, 3, 2, 17, 1, 13, 1, (2) 53, 19, 41, 25, 13, 20, 31, 4, 65, 5, 52, (19)	6175 3107 1 14243
1353k	36; 1, 7, 2, (2) 54, 18, 50, 9, 2, 29, (25)	42191 8628453336
1353l	36; (10) 55, 3, 17, 1, 23, (7) 1, (17)	— —
1353m	36; 56, 16, 3, 49, (4)	—
1353n	36; 1, 1, 1, 2, 1, 8, (2) 31, 32, 37, 21, 49, 8, (33) 57, 3, 39, —	— — —

TABLE 1001 TO 1500—CONTINUED.

No.	Continued-fraction lines	y / x
1354*	36; 58, 14, 1, 5, 15, 1, 2, 2, 6, 1, 7, 3, 4, 1, 1, (2, 2)	131355 7459053930 9948611635
1355	59, 33, 10, 57, 9, 22, 15, 38, 33, (25, 25)	918484293523829
1356	—	—
1357	2, (14)	4918739790
1358	25, 26, (5)	5212
1359	—	1 9196229
1360	—	—
1361*	36; 60, 13, 1, 41, 8, 2, 1,	9 65303 46430 64640
1362	1, 11, —	355 59347 94143 8721345014
1363	5, 1, 4, (18)	—
1364	—	—
1365	61, 12, 44, 8, 57, 11, 52, 15, (4) 1, 6, 1, 7, 1, 1, 1, 5, 2, 17, 1, 23, 1, 1, 1, 1, (2) 62, 11, 22, 37, 9, 44, 27, 43, 12, 33, 4, 67, 3, 44, 29, 33, 36, (23)	— — —
1365b	36; 1, 36, 2, (36)	—
1365c	36; 63, 10, 27, 47, 51, 15, 8, 34, 35, 25, 23, 45, 14, 37, 35, (18)	468 99823 17281 4788023410
1365d	36; 64, 1, 1, 17, (3)	85769 2 7749
1365e	36; 65, 7, 4, 41, 9, 6, (16)	58529
1365f	36; 9, 1, 4, 2, 1, 2, 1, 1, 14, 5, 1, (1, 1) 25, 40, 23, 32, 41, 5, 13, 40, (31, 31)	35 93384 1325 66186 45260
1365g	36; 66, 11, 6, 4, 1, 23, 3, 1, 4, 2, (36)	13730711 2847205478
1365h	36; 67, 7, 39, 17, 49, 14, 33, (2)	—
1365i	36; 1, 9, 1, 1, 3, 34, (1) 1, 68, 17, 16, (58) 1, 2, (6)	9531479265 1062 2867575310

No.	Continued-fraction lines	y / x
	7, 43, 25, (11)	34
1365j	69, 31, — 35, 55, 9, (35) 6, 4	684 25271
1365k 1366	36; 4, 10, 1, 7, 1, 1, 4, 2, 1, 1, 14, 5, 4, 1, 2, 1, 2, 2, 70, 5, 9, 38, 35, 15, 27, 31, 42, 5, 14, 15, 46, 21, 42, 23, 30, —	— 61 98787 91120 98468 23128 64853 37 2291 03705 28461 89335 14238 95408 6404327
1367 1368	3, 17, 1, 3, 2, 1, 1, 1, 3, 3, 1, 4, 1, 11, 2, (36) 1, 25, 37, 30, 39, 19, 18, 49, 13, 57, 6, 35, (2)	— —
1370* 1371 1372	36; (35)	99525 1368 37 1
1373*	36; (1) (72)	371
	37; (74)	1370
	37; (37) 24, 24,	—
	37; 1, 1, 8, 2, 1, 1, 1, 2, 2, 5, 1, 3, (18) 7, 19, 58, 27, 36, 31, 37, 24, 29, 12, 49, 19, (4) 49, —	92062 30469 2374156527 34 10035 48679 685
	37; 14, 43, (37, 37) (21)	25382
	37; 4, 2, 1, 3, 4, 1, 2, (24) 57, 9, 42, 19, 15, 43, 26, (3) 5, — (1 ...)	41422 8295007445 15 35443 —

TABLE 1001 TO 1500—CONTINUED.

No.	Continued-fraction lines	y / x
1375 1376	37; 12, 2, 1, 7, 1, 1, 3, 2, 1, 2, (6) 6, 25, 46, 9, 39, 34, 19, 25, 39, 26, (11)	24090 73932 8 93308 52249
1376b	37; 10, 1, 1, 2, 2, 3, 1, (17) 7, 41, 32, 25, 28, 17, 55, (4)	58 37205 2165 28049 11121 62480 4 12700 701410513
1377 1378*	37; 9, 3, 1, 3, 1, 7, 2, 5, (4) 8, 19, 47, 16, 53, 9, 32, 13, (17)	42801 678 06016 25179 68895
1378b	37; 8, (4, 4) 9, (17, 17)	182 6761
1379	37; 7, 2, 2, 2, 1, 1, 3, 3, (10) 10, 29, 26, 25, 34, 37, 19, 22, (7)	— —
1380	37; 6, 1, (2) 11, 49, (20)	— —
1380b	37; 6, 5, 1, 1, 4, 2, 2, 3, 3, 4, 14, 1, 1, 1, 2, 1, 1, 2, 1, 24, 12, 13, 37, 36, 15, 28, 27, 20, 21, 17, 5, 45, 28, 39, 23, 35, 36, 21, 52, 3,	— —
1381*	18, 1, 1, 5, 1, 2, 7, 1, 9, 1, 2, 1, (4, 4) 4, 39, 35, 12, 45, 25, 9, 60, 7, 51, 20, 47, (15, 15)	5 76794 95050 76394 67609 26124 87905 214 34743 40923 77992 50381 97708 76282
1382	37; 5, 1, 2, 2, 1, 1, (36) 13, 46, 23, 26, 31, 43, (2)	1303 4039278327 12 9344984426815
1383	37; 5, 3, 2, 1, (11) 14, 21, 23, 49, (6)	17299 2797 1 04092
1384 1385* 1386	37; 4, 1, (17) 15, 57, (4) —	21259752 40557 15 10442
1386b	37; 4, 1, 1, 1, (3, 3) 16, 41, 29, 40, (19, 19)	2 20986 97686 82 33062 52807
1386c	37; 4, 2, 1, (2)	16796

No.	Continued-fraction lines	y / x
	17, 25, 41, (22)	6 25975 6165 38516 07582 2 29862 22854 32981
1387	37; 4, 8, (37) 18, 9, (2)	90 81305 156472
1388	37; 3, 1, 9, 1, 8, 2, 2, 5, 3, (18)	—
1389	19, 52, 7, 61, 8, 29, 28, 13, 23, (4)	—
1389b	37; 3, 1, 2, 2, (24) 20, 43, 23, 31, (3)	— —
1390	37; 3, 1, 1, 6, 4, 1, 4, 1, 1, 11, 1, 7, 2, 1, 2, 1, (6)	—
1391	21, 34, 39, 11, 15, 51, 14, 35, 39, 6, 61, 9, 26, 41, 21, 49, (10)	—
1392	12, 25, 38, 29, 43, 10, 55, (13)	—
1392b	37; 3, (4) 23, (16)	— —
1393	37; (34)	96
1394	(1, (07)) 2, 1, (36) 25, 49, (2)	3538336 12545 —

TABLE 1001 TO 1500—CONTINUED.

No.	Continued-fraction lines	y / x
1395	37; 26, 2, 11, 1, (1, (89))	45 3320
1396	—	1
1397	20, 1, —	24001
1398	—	—
1399	27, 45, 3, 5, 48, 25, (1, (84))	169964840530
1400	—	—
1401	28, 37, 2, 1, 29, 14, 1, —	4289 48201
1402*	37; 2, 37, 24, 41, 6, 17, 18, 1, 1, 2, (6)	66440
1403	4, 43, 31, 28, (11)	—
1404	37; 1, (36)	1 6033428 48351
1405	—	—
1406	12, 3, 2, 51, (2)	1 3606145347
1407	— 23, 38, 14, 1, 7, 2, 1, 1, 1, 4, 2, 1, 3, 2, 7, 24, — 37; 2, 1, 4, (37)	— 19 32130 14430 04420 15349 449 722 67866 47899 40453 09640
	30, 11, 14, 5, 62, 9, 27, 37, 30, 41, 15, 25, 43, 18, 31, 10, 3, 58, 15, (2) 37; 2, — (25) 37; 2, 21, 1, 2, 5, 2, 2, 1, 1, 6, 4, 1, 1, 8, 1, 4, 10, 2, (24) 32, 17, 5, 49, 24, 13, 29, 25, 33, 40, 11, 16, 35, 39, 8, 55, 15, 7, 35, (3)	— 12 — 44183 34363 33784 16 53779 66015 12296 9857091352 2 48379 80029 00509 93 00157 1995
	37; 2, 3, 11, 1, 1, 2, 1, 7, 1, (1, 1) 79, 63, 1, 6, 41, 33, 22, 49, 9, 42, (31, 31) 33, 13, 19, 3, 54, 5, (1)	— — —
	34, 3, (46) 37; 2, 7, 1, (4) 35, 9, 56, (13)	74176266 62425 — 2249
	37; 2, (1, (45)) 38, 36,	602 100 01357257915

No.	Continued-fraction lines	y / x
	37; (37)	3211
		1 20487 4967231777
	37; 1, (26, (43)) 37, 1,	94474 40193 53380
		35 46252 44206
	16, 39, 1, 2, 7, 1, (17)	20
1408	—	118 21953
1409*	11, 33, 28, 9, 63, (4)	4440
1410	—	71373501
1411	39, 36, 7, 3, 9,	1 012577013 014535932
1412	—	124 23214 3310874490
1413	23, 8, —	4669 87287
1414	—	—
1415	40, 35, 5, 64, 7, 55, 16, 49, 17, 40, 31, 35, 32, 37, (25, 25)	485 26483 4121289190
	37; 1, (4)	18247 50630
	(19, 1, 1, 2)	2205 2895060241
	41, 34, 33, 23, 30, 15, 38, 35, 21, (2)	—
	37; 42, 1, 1, 1, 3, 4, 6, 1, 1, 2, (18)	—
	43, 32, 1, 1, 44, 19, 17, 11, 41, 32, 29, (4)	—
	44, 1, 3, 1, 6, 18, 1, 1, 1, 5, (8)	—
	19, 36, 37, 17, 52, 11, 4, 47, 27, 44, 13, (9)	—
	45, 29, 30, 31, 1, 11, 1, 14, 8, 3, 2, 5, 2, 1, (4)	—
	43, 6, 65, 5, 9, 21, 30, 13, 25, 45, (14)	—
	41, 14, 49, 19, (10)	—
	46, 35, 34, 31, —	—

TABLE 1001 TO 1500—CONTINUED.

No.	Continued-fraction lines	y / x
1416	37; 1, 28, 25, 2, 1, (8)	6858
1417*	47, 1, 39, 16, 23, 49, (8)	2 58065
	1, 27, 24, 1, 7, (2, 2)	435
	48, 43, 3, 64, 9, (29, 29)	72829
	26, 47, 2, (61, (07, 7))	16402 14636
1418*	—	381803933939
1419	49, 25, —	—
1420	(3, (72))	—
1421	—	12770
1422	—	—
1423	37; 50, 24, 12, 22, 36, 39, (14)	2554383089
1424	—	9 260
1425	37; 51, 23, —	—
1426	—	31
1427	(2, (92))	1199
1428	—	—
	52, 2, 18, 29, —	9801
1429*	—	08097
1430	37; 1, 1, 1, 3, 5, (8)	79792
1431	—	127 24746 01715 56263 0946270589
1432	3, 47, 19, 14, (9)	4800 11423 18614 76037
	53, —	13250
1433*	—	5 00001
1434	21, —	1514
	54, 20, 17, 38, 33, 17, 31, 42, 11, 18, 3, 66, 7, 57, 14, 33, 6, 9, 22, 21, 19, 42, 27, (2)	—
	37; 1, 2, (4)	—
	1, 44, 25, (16)	—
	55, 47, 3, —	—

No.	Continued-fraction lines	$y \ / \ x$
	37; (2) (19)	—
	37; 56, 18, 15, 2, 3, 1, 1, 1, (2)	2 601917352
		98 47097
	37; 57, 17, 41, 13, 17, 41, 50, 39, (23)	2481
		33
		93739 18762
	58, 3, 16, 31, 26, 41, 23, 37, 34, 29, (2)	340910
	37; 1, (2)	89305
	65, 9, 13, 5, 14, 87, (212)	75918
	37; (2, (32, 232))	1333153125
	61, 14, 4, 29, 51,	—
	37; (2)	—
	62, 13, 4, 4, (26)	—
	37; (1)	—
	4, (5, 14)	1324
	63, 12, 23, 1, 8, 10, 1, 1, 2, 4, (9)	17 6456769682 0656739673
		53237
	37; 63, 9, 7, 49, 24, 17, (8)	—
		1523
		57673 29724
		46653 69200
	64, 11, 5, 59, 8, 43, 31, 32, 41, 13, 16, (37, 37)	17 66690
		2968377531
	37; 65, 10, 21, 2, 1, 1, 1, 2, 1, 4, 3, (12)	9
	26, 25, 38, 31, 39, 22, 47, 15, 23, (6)	99046
	37; 66, 1, 88, 2, 1, 33, (7)	2 596189490
	7, 31, 21, —	—
1436	37; 67, 9, 2, (18)	—
	35, (4)	—

TABLE 1001 TO 1500—CONTINUED.

No.	Continued-fraction lines	$y \ / \ x$
	68, —	256
1437	37; 1, 71, 1, 5, 2, 2, 4, 18, 1, 2, 1, 1, 1, (24)	42 83205 10399
1438	2, 29, 28, 17, 4, 53, 21, 41, 28, 47, (3)	64040
1439	69, 1, —	—
1440	14, 1, 1, 7, 1, 3, 3, 24, 1, (36)	91753956 903839 1849254806064
	37; 1, 9, 9, 1, 27, 47, 46, 9, 53, 18, 23, 3, 71, (2)	—
		07281 221757241139
	37; 2, 1, 5, 6, 1, 2, 1, 1, 2, 2, 1, 10, 7, 2, (37)	338 95537 37610 64960
	70, (17) 5, 25, 46, 13, 11, 49, 22, 37, 35, 26, 23, 50, 7, 10, 35, (2)	12857 98517 00303
	37; 1, (4)	—
1441	37; 9, 2, 1, 2, 2, 1, 3, 1, 3, 4, 1, 3, 1, 14, 2, 1, 1, 4, 2, (6)	221 54985 95953 3851124040 54999
1442	43, 8, 27, 40, 25, 24, 45, 17, 48, 19, 15, 51, 16, 57, 5, 29, 33, 40, 15, 32, (11)	8410 14472 84306 8270308
1443	—	—
	4, (36)	721
1445*	37; 1, —	—
1446	—	1443
1447	—	—
1448	73, (2)	1
		38
	37; (74)	381
		38
	38; (7, (61)) 2,	—

No.	Continued-fraction lines	y / x
	38; (25, (38)) 26, — 38; 7, 1, 3, 1, 1, 2, 3, 1, 5, 12, 1, 1, (37) 9, 54, 17, 39, 34, 27, 18, 51, 13, 6, 37, 39, (2) 38; (3, 4) 47, 1, 2, —	144159 3 42829 14389 10093 130 41033 17574 21848 — — —
1449 1450* 1451 1452 1453* 1454 1455 1456 1457	38; 1, 6, 6, 1, 7, 2, 6, 4, (8) 12, 25, 25, 17, (9) — 38; 10, 5, 1, 45, (8, 8) 38; 8, 4, 69, 1, 1, (9, 9) 10, 1, 1, 2, 2, 3, 1, 1, 2, 2, 14, 1, 4, 1, 1, (37) — — — 8, 2, 1, 49, 23, 29, 19, 38, 35, 25, 31, 5, 59, 14, 35, 41, (2)	2773 0814284090 723 1 0811943933 — 31566 20707 85340 38 28530 — — — 32 50782 7844732
	38; 7, (1, (84))	2 626088778
	38; 37, 39, 9, 1, 1, 5, 1, 3, 6, 10, 1, 2, 1, 2, 1, 1, 3, 18, 1, 3, 1, 1, 9, 31, 33, 41, 12, 51, 19, 12, 7, 52, 21, 44, 23, 36, 37, 21, 4, 57, 17, 36, 3, 1, 1, 2, 6, 2, 1, 2, (25, 25) 33, 28, 1, 27, 39, 28, (3, 3) 38; 10, 6, 1, 9, 2, 3, 1, 4, 1, 1, 2, 2, 4, 15, (38) 11, 67, 46, 31, 6, 37, 25, 29, 17, 5, (2) 38; 1, 25, (6) 38; 12, 11, (17, 1) 48, 9, (3) 38; 13, 5, 9, 55, (16) 8, 29, 32, 43, 7, 23, 16, (31) 56, —	— 54805 37 58165 73334 54563 1432 54652 50389 3444326 9776194930 96169 965981908898245 6707479 23766243 1 29855 18 54951 9341965238 3160730132 — 723

TABLE 1001 TO 1500—CONTINUED.

No.	Continued-fraction lines	y / x
1458 1459 1460 1461 1462 1463 1464 1465* 1466 1467 1468 1469 1470 1471 1472 1473 1474 1475 1476	38; 5, 18, 1, 3, 1, 2, 1, 1, 7, 1, 10, (38) — — 38; 1, 49, 17, 26, 23, 34, 41, 9, 62, 7, (2) 13, 2, 1, 1, 7, 8 — 6, 39, 3, 37, 46, 1, 4, 2, 2, 2, 1, 1, 1, 5, 4, 15, 25, 2, 1, 1, (37) 45, 15, 29, 27, 25, 39, 30, 43, 13, 18, 5, 3, 30, 31, 45, (2) — — — 18, 64, 2, 19, 16, 35, 4, 1, (1, (84)) — — 18, 74, 49, 14, 1, 5, 3, 2, 1, 3, 3, 18, 1, 4, 6, 1, 2, 1, (24) 41, — — 3, 61, 12, 20, 25, 44, 19, 23, 4, 59, 15, 11, 51, 20, 55, (3) — 1884, 33, 5, 31, —	88890 25540 27540 22872 25694 24820 3012 96401 78 52281 — 81439 56337 9998412034 45032 81010 9 239 25898 — — — 6 15529 10897 33506 4 16529 16197 65845 0827252785 — 97289 — 2753

No.	Continued-fraction lines	$y \ / \ x$
1477	(-74)	—
1478	—	—
	1894, 1, —	4
	2308, 1, 15, 2, 1,	23712593
	—	72
	25, 47, (1, (25))	9 07925
	—	14427
	1383, 49, 2, 4, 1, 2, 1, 1, 4, 4, 1, 7, 1, 2, (3, 3)	2 0726490 21285
		82 84932
	39, 16, 15, 56, 9, 49, 24, (21, 21)	—
	1382, 40, 31, 6, 39, 24, 17, 4, 26, 2, 35, 2, (2, 2)	1533 05185
	11, 17, (25, 25)	16
		52043
	1383, 1, 23, 1, (3, (72))	73045
	46, 27, 41,	80026
		640
	2843, 1, 13, 1, 1, 3, 49, 25, 9, 1, 1, 5, 1, 6, 8, 2, 1, 2, 1, 1,	18883 73036 35216
	(18)	
	18, 36, 19, 3, 8, 39, 37, 12, 57, 11, 9, 27, 41, 24, 33, 43, (4)	46281 85247
		64922
		7 23520 34792
	(1, (34)) 853, 5, 369, 33,	19836
		7 60216358
	2682, (11) 4, (5)	—
	49, —	5291
	328, 1, 4, 1, 4, 3, 2, 4, 12, 1, 1, 3, 1, 2, 1, 6, 1, 14, 2, 8, 25,	36737
	17, 6, 41, 35, 18, 47, 21, 51, 10, 63, 5, 35, 9, 3,	66160
	27, 45, 1, 14, 53, 15, 21, 30, 1, 1, 10, 2, 1, (37)	89163 97276 23871 56715
	30, 45, 7, 25, 51, (2)	1 01712 37347 28288 54875 60183 30
	28, 1, 34, 1, 15, 42, 2, 4, 1, 1, 3, 17, 1,	39 01057
	38; 2, (2)	1151
	2922, 41, (2, 3) 1, 2, 2, 2, 9, 5, 1, 3, 1, (24)	2656 14220
	24, 31, 8, 13, 53, 16, 59, (3)	—
	39, 23, 43, 144, 57, 7, 1, 2, 2, 4, 1, 2, 3, 1,	8962979326 93399
	38; 2, 37, 6, 9, 1, 50, 23, 30, 15, 46, 25, 18, 55, 10, 5, (38)	1 43151 08109
	3082, 1, 1, 7, 15, (2)	95957 60507 79476
	2, 32, (2, (52))	29745
	2393, 41, 11, —	54 6114769298
	41, 34, —	5
		83
	3812, 25, 6, 1, 2,	2 18560
	20, (9)	—
	37, 23, 3, (8)	—
	3126, 8, 2,	—
	3282, 21, 415, 1, 1, 1, 1, 5, 1, 3, 1, 2, 18, 1, 6, 25, 2, (10)	2 61633 47062 2926284001
	9, 28, 37, 33, 36, 31, 43, 12, 53, 17, 44, 27, 4, 63, 11, 3, 36,	100 55043 41504 36351
	(7)	
	17, 3,	728102117
	3832, (38)	1 193711174
	34, 4, (2)	59

TABLE 1001 TO 1500—CONTINUED.

No.	Continued-fraction lines	$y \ / \ x$
1479	38; 2, 5, 2, 2, 1, 1, 1, (1)	1 14525
1480	35, 13, 30, 25, 38, 33, 35, (34)	44 04376
1481*	—	948 5559

No.	Continued-fraction lines	y / x
1482	38; 2, 8, (19)	—
1483	36, 9, (4)	2 13859
1484	—	—
1485	38; 2, 14, 1, 8, 1, 2, 5, 1, 1, 2, 1, 4, 10, 1, 3, 1, 1, (1, 1)	1 01945 97513 61405
1486	37, 5, 65, 8, 49, 25, 13, 40, 35, 23, 47, 16, 7, 56, 17, 41, 32, (35, 35)	39 23264 72655 17468
1487	—	—
	38; (2) (38)	2 77
	38; 1, 1, 25, 5, 1, 7, 1, 2, 1, 1, 1, 1, 2, 4, 6, 1, 3, 2, 2, 1, 1, 12, 3, 1, (37)	41 20928 70570 44632 38045
	39, 38, 3, 13, 58, 9, 51, 22, 39, 33, 34, 37, 27, 17, 1, 53, 18, 29, 26, 33, 43, 6, 19, 57, (2)	1586 95889 00942 30191 81226
	38; 1, 1, (10)	44
	40, 37, (?)	1695 1079 28
	38; 1, 1, (6)	—
	41, 36, (1)	—
	38; 1, 1, 4, 1, 1, 1, 3, (38)	4 92228
	42, 35, 15, 43, 30, 41, 21, (2)	189 74735 2 05495 79 24232145
	38; 1, 1, 3, 1, 1, 3, 1, (37)	—
	43, 34, 19, 37, 38, 17, 59, (2)	—
1488	44, 33, 23, 49, (1)	12151 25 78145 994 84332
1489*	38; 1, 1, 2, 2, 1, 4, 2, 3, 1, (1, 1)	5
1490*	45, 32, 27, 24, 47, 15, 31, 19, 40, (33, 33)	1716973
1491	—	—
1491b	38; 1, (1, 1)	68230
	46, (31, 30)	13 56270 523 87849
	38; 1, 1, 1, 1, 2, 2, (1)	—
	47, 30, 35, 37, 26, 25, (42)	—
1492	38; 1, 1, 1, 2, 10, 1, 1, 1, (18)	—
	48, 29, 39, 28, 7, 48, 27, 49, (4)	—
1493*	38; 1, 1, 1, 3, 2, 2, 18, 1, (10, 10)	1694 41225
1494	49, 28, 43, 19, 28, 31, 4, 67, (7, 7)	65471 00182
	38; 1, 1, 1, 7, 15, 5, (38)	17040490 0052251645
	50, 27, 47, 10, 5, 25, (2)	118673
1495	38; 1, (1)	3365
	51, (26)	—
1496	38; 1, 2, (9)	—
	52, 25, (8)	—
1497	38; 1, 2, 4, 4, 1, 1, 1, 1, 6, 2, 2, 1, 8, 1, (24)	58 19614 30932
1498	53, 24, 17, 16, 41, 33, 32, 43, 11, 31, 23, 51, 8, 67, (3)	2251 67187 51127
1499	38; 1, 2, 2, 1, 1, 1, 3, 1, 12, 8, 1, 1, (10)	83118 18384
1500	54, 23, 26, 39, 31, 42, 17, 57, 6, 9, 38, 39, (?)	32 17006 59967
	38; 1, 2, 1, 1, 7, 5, 1, 4, 1, 2, 3, 1, 2, 1, 1, 2, 10, 1, 2, 15, 6, 1, (37)	77552 05839 43067 80299
	55, 22, 35, 41, 10, 13, 55, 14, 47, 25, 19, 46, 23, 38, 35, 29, 7, 49, 26, 5, 11, 65, (2)	30 02576 94656 26953 60610
	38; 1, 2, 1, 2, 2, 1, 6, 2, 1, (18)	118 11844
	56, 21, 44, 25, 24, 49, 11, 25, 51, (4)	4574 70751

In connexion with the subject we have a paper, “A Table of the Square Roots of Prime Numbers of the form

$4m + 1$ less than 10000 expanded as Periodic Continued Fractions,” by C. A. Roberts, with Introduction and Explanation by Artemas Martin, the *Mathematical Magazine*, vol. II. (No. 7, for October, 1892), pp. 105–120. This extends, in fact, to numbers up to 10501, but only the denominators of the continued fractions (that is, the first lines of Degen’s and the present table) are given: thus the entry for 1009 is 31; 1, (3, 3).

The paper just referred to notices errors in Degen’s tables for the numbers 853 and 929. For 853 the first line should be

$$29, 4, 1, 5, 1, 2, 4, 1, 1, 15, 19, (2, 2),$$

(15 instead of Degen’s 14). For 929 the first and second lines should be

$$30, 2, 11, 1, 2, 3, 2, 7, 5, (2, 2)$$

$$1, 29, 5, 40, 19, 16, 25, 8, 11, (23, 23).$$

The values of x, y in Table I. and those in Table II. (for the solution of $y^2 = ax^2 - 1$) are correct for each of the numbers 853 and 929.

ON HALPHEN'S CHARACTERISTIC n , IN THE THEORY OF CURVES IN SPACE.

[949]

[From *Crelle's Journal d. Mathematik*, t. CXI. (1893), pp. 347–352.]

If we consider a curve in space without actual singularities, of the order (or degree) d , then this has a number h of apparent double points (adps.), viz. taking as vertex an arbitrary point in space, we have through the curve a cone of the order d , with h nodal lines; and Halphen denotes by n the order of the cone of lowest order which passes through these h lines. For a given value of d , h is at most $= \frac{1}{2}(d-1)(d-2)$, and as shown by Halphen it is at least $= [\frac{1}{4}(d-1)^2]$, if we denote in this manner the integer part of $\frac{1}{4}(d-1)^2$. For given values of d , h , it is easy to see that n must lie within certain limits, viz. if ν be the smallest number such that $\frac{1}{2}\nu(\nu+3)$ equal to or greater than h , then n is at most $= \nu$; and moreover n must have a value such that nd is at least $= 2h$, or say we must have $nd = 2h + \theta$, where θ is $= 0$ or positive. For any given value of d , we thus have a finite number of forms (d, h, n) , and we have thus *primâ facie* curves in space of the several forms (d, h, n) : but it may very well be, and in fact Halphen finds, that when $d = 9$ or upwards, then for certain values of h , n as above, there is not any curve (d, h, n) ; thus $d = 9$, $h = 17$ the values of n are $n = 4$, $n = 5$, but there is not any curve $d = 9$, $h = 17$ for either of these values of n ; or say the curves $(9, 17, 4)$ and $(9, 17, 5)$ are non-existent. And in the Notes and References to the papers 302, 305 in vol. v. of my *Collected Mathematical Papers*, 4to. Cambridge, 1892, see p. 615, I remarked that, in certain cases for which Halphen finds a curve (d, h, n) , such curve does not exist except for special configurations of the h nodal lines not determined by the mere definition of n as the order of the cone of lowest order which passes through the h nodal lines; for instance $d = 9$, $h = 16$, $n = 4$, for which Halphen gives a curve, I find that for the existence of the curve it is not enough that the 16 lines are situate upon a quartic cone, but they must be the 16 lines of intersection of two quartic cones.

In fact, starting from an existing curve, say the complete intersection of two given surfaces of the orders μ , ν respectively, we have, as is known,

$$d = \mu\nu, \quad 2h = \mu\nu(\mu-1)(\nu-1) :$$

we find also, as will appear, $n = (\mu-1)(\nu-1)$: hence also $nd = 2h$, viz. the cone of the order n through the h nodal lines meets the cone of the order d in these lines counting each twice, and in no other lines. I remark also that h is $= \frac{1}{2}n(n+3)$ if $\mu + \nu = 4$; viz. we have here two nodal lines lying in a plane; but if $\mu + \nu > 4$, then h is greater than $\frac{1}{2}n(n+3)$, viz. in this case the nodal lines are not h lines taken at pleasure, but they are lines subject to the condition of lying in a cone of the order n . But this is not all; the h nodal lines lie not only in this cone of lowest order n , but also in cones of the orders $n+1$, $n+2$, $\dots$, $n + (\mu + \nu - 2)$ respectively: I do not for the moment attempt to determine the number of independent conditions which are hereby imposed upon the h lines.

The last-mentioned theorem constitutes in fact the geometrical interpretation of results contained in Jacobi's paper "De eliminatione variabilis e duabus aequationibus algebraicis," *Crelle*, t. xv. (1836), pp. 101–124, [*Ges. Werke*, t. III. pp. 295–320]. Consider the two equations

$$U = (*|x, y, z, w)^\mu = 0; \quad V = (*|x, y, z, w)^\nu = 0;$$

representing surfaces of the orders μ , ν respectively; since the form of the equations is quite arbitrary, we may without loss of generality assume that the point

$$(x, y, z) = (0, 0, 0)$$

is an arbitrary point in space; and this being so, we find the equation of the cone, vertex this point, which passes through the curve of intersection of the two surfaces by the mere elimination of w between the two equations. As the reasoning is exactly the same for a particular case, I write for convenience $\mu = 3$, $\nu = 4$, and consider the two equations

$$A_0w^3 + A_1w^2 + A_2w + A_3 = 0, \quad B_0w^4 + B_1w^3 + B_2w^2 + B_3w + B_4 = 0,$$

where the suffixes show the degrees in regard to (x, y, z) , viz. A_0 , B_0 are mere constants, A_1 , B_1 are linear functions $(*|x, y, z)^1$, A_2 , B_2 quadric functions $(*|x, y, z)^2$, and so on. Multiplying the first equation

successively by $1, w, w^2, w^3$, and the second equation successively by $1, w, w^2$, we have 7 equations from which to eliminate $1, w, w^2, w^3, w^4, w^5, w^6$, and the result is

$$\begin{vmatrix} A_0 & A_1 & A_2 & A_3 & \cdot & \cdot & \cdot \\ \cdot & A_0 & A_1 & A_2 & A_3 & \cdot & \cdot \\ \cdot & \cdot & A_0 & A_1 & A_2 & A_3 & \cdot \\ \cdot & \cdot & \cdot & A_0 & A_1 & A_2 & A_3 \\ B_0 & B_1 & B_2 & B_3 & B_4 & \cdot & \cdot \\ \cdot & B_0 & B_1 & B_2 & B_3 & B_4 & \cdot \\ \cdot & \cdot & B_0 & B_1 & B_2 & B_3 & B_4 \end{vmatrix} = 0,$$

viz. this is an equation

$$(*|x, y, z)^{12} = 0,$$

of the cone of the order $d = 12$, through the curve of intersection of the two surfaces: say this equation is $\Omega = 0$.

But if we only multiply the first equation by $1, w, w^2$ successively and the second equation by $1, w$ successively, then, we have 5 equations serving to determine the ratios of $w^5, w^4, w^3, w^2, w, 1$, viz. we have these quantities proportional to the six determinants which can be formed out of the matrix

$$\begin{vmatrix} A_0 & A_1 & A_2 & A_3 & \cdot & \cdot \\ \cdot & A_0 & A_1 & A_2 & A_3 & \cdot \\ \cdot & \cdot & A_0 & A_1 & A_2 & A_3 \\ B_0 & B_1 & B_2 & B_3 & B_4 & \cdot \\ \cdot & B_0 & B_1 & B_2 & B_3 & B_4 \end{vmatrix},$$

say we have

$$w^5 : w^4 : w^3 : w^2 : w : 1 = L : M : N : P : Q : R,$$

where L, M, N, P, Q, R represent homogeneous functions $(*|x, y, z)^e$, of the degrees 11, 10, 9, 8, 7, 6 respectively. We may if we please write

$$\frac{w^5}{L} = \frac{w^4}{M} = \frac{w^3}{N} = \frac{w^2}{P} = \frac{w}{Q} = \frac{1}{R},$$

or eliminating w , we have the series of equations which may be written

$$\begin{vmatrix} L & M & N & P & Q \\ M & N & P & Q & R \end{vmatrix} = 0,$$

viz. we thus denote that the determinants formed with any two columns of this matrix are severally $= 0$. This of course implies that each of the determinants in question is the product of Ω and a factor which

is a homogeneous function of the proper degree in (x, y, z) , so that the several equations are all of them satisfied if only $\Omega = 0$. We have for instance $PR - Q^2 = A\Omega$, where A is a quadric function $(*|x, y, z)^2$; similarly $NR - PQ = B\Omega$, where B is a cubic function $(*|x, y, z)^3$; and the like as regards the other determinants.

If the ratios $x : y : z$ have any given values such that we have for these $\Omega = 0$, then w has a determinate value, that is, on each line of the cone $\Omega = 0$, there is a single point of the curve of intersection of the two surfaces: the only exceptions are when, for the given values of $x : y : z$, the expressions for w assume an indeterminate form, viz. w has then two values, and there are upon the line two points of the curve, or what is the same thing, the line is a nodal line of the cone: the conditions for a nodal line thus are $L = 0$, $M = 0$, $N = 0$, $P = 0$, $Q = 0$, $R = 0$, viz. each of these equations is that of a cone passing through the nodal lines of the cone $\Omega = 0$; the cone of lowest order is $R = 0$, a cone of the order 6 meeting the

cone $\Omega = 0$ of the order 12 in 36 lines each twice, which lines are consequently the nodal lines of the cone $\Omega = 0$. The mere condition of the 36 lines lying upon a cone of the order 6 shows that the 36 lines are not arbitrary; and we have moreover, through the 36 lines, cones of the orders 7, 8, 9, 10 and 11 respectively. Obviously the foregoing reasoning is quite general, and for the surfaces of the orders μ, ν we have, as stated above, the cone Ω of the order $\mu\nu$, with $h = \frac{1}{2}\mu\nu(\mu-1)(\nu-1)$ nodal lines, the intersections (each counting twice) of this cone with a cone of the order $n = (\mu-1)(\nu-1)$; and moreover the h nodal lines lie also in cones of the orders $n+1, n+2, \dots, n+\mu+\nu-2$ respectively.

To examine the meaning of the theorem, I form the table

μ, ν	d	h	$n, \dots, n + \mu + \nu - 2$
2, 2	4	2	1, 2, 3
2, 3	6	6	2, 3, 4, 5
2, 4	8	12	3, 4, 5, 6, 7
2, 5	10	20	4, 5, 6, 7, 8, 9
$\vdots$			
3, 3	9	18	4, 5, 6, 7, 8
3, 4	12	36	6, 7, 8, 9, 10, 11
3, 5	15	60	8, 9, 10, 11, 12, 13, 14
$\vdots$			
4, 4	16	72	9, 10, 11, 12, 13, 14, 15
4, 5	20	120	12, 13, 14, 15, 16, 17, 18, 19

Here $\mu, \nu = 2, 2$, there are 2 nodal lines, which are arbitrary, and of course lie on cones of the orders 1, 2, 3 respectively. So $\mu, \nu = 2, 3$; there are 6 nodal lines, which are not arbitrary, inasmuch as they lie on a cone of the order 2; but regarding them as arbitrary lines on such a cone, we can through them draw a cone of the order 3 or any higher order, and it is thus no specialisation to say that they lie upon cones of the orders 3, 4, and 5. But going a step further $\mu = 2, \nu = 4$: here we have 12 nodal lines which, inasmuch as they lie on a cone of the order 3, are not arbitrary: and they are not arbitrary lines upon this cone, for they lie on a cone of the order 4, and such a cone can be drawn through at most 11 arbitrary lines on a cubic cone. In fact, upon a cone of the order θ , taking at pleasure N lines, the condition that it may be possible through these to draw a cone of the order $\theta + 1$ is $\frac{1}{2}(\theta + 1)(\theta + 4) = N + 3$ at least; for if this number were $= N + 2$, then through the N points we have only the improper cone $(\alpha x + \beta y + \gamma z) U_\theta = 0$, if $U_\theta = 0$ is the cone of the order θ . It thus appears that the 12 nodal lines are not arbitrary lines on a cubic cone, but that they constitute the complete intersection of a cubic cone and

a quartic cone. But through such 12 lines we may draw cones of the 5th and higher orders, and it is thus no further condition that the 12 lines lie on cones of the orders 5, 6 and 7 respectively.

So again $\mu = 3, \nu = 3$; we have here 18 nodal lines which, inasmuch as they lie on a cone of the order 4, are not arbitrary: and they are not arbitrary lines on this cone inasmuch as they lie also on a cone of the order 5, and such a cone can be drawn through at most 17 arbitrary lines on the quartic cone: it thus appears that the 18 nodal lines are 18 out of the 20 lines of intersection of a quartic cone and a quintic cone. But there is no further condition, for through such lines we can draw a cone of the order 6 or any higher order

and thus the lines lie on cones of the orders 6, 7 and 8 respectively. It appears probable however that, for higher values of μ, ν , it would be necessary to take account not only (as in these examples) of the cones of the orders n and $n + 1$, but of those of higher orders $n + 2$, &c.; and thus that it is not the true form of the theorem to say that the h nodal lines must be h out of the $n(n + 1)$ lines of intersection of two cones of the orders n and $n + 1$ respectively.

It appears, by what precedes, that the $h, = \frac{1}{2}\mu\nu(\mu - 1)(\nu - 1)$, lines which are the nodal lines of the cone of arbitrary vertex which passes through the curve of intersection of two surfaces of the orders μ, ν respectively, form a remarkable special system of lines, which well deserve further study. I remark also that, without having proved the negative, it seems to me clear that given the values of d, h, n it is only in the cases where the h lines form some such special system (and not in the general case where the h nodal lines are any lines whatever on a cone of the order n) that there exists a curve (d, h, n) ; and thus that the question for further investigation is, for given values of (d, h, n) to determine the conditions necessary for the existence of a curve in space with these characteristics (d, h, n) .

ON THE SEXTIC RESOLVENT EQUATIONS OF JACOBI AND KRONECKER.

[950]

[From *Crelle's Journal d. Mathematik*, t. CXIII. (1894), pp. 42–49.]

The equations referred to are: the first of them, that given by Jacobi in the paper “Observatiunculae ad theoriam aequationum pertinentes,” *Crelle*, t. XIII. (1835), pp. 340–352, [*Ges. Werke*, t. III., pp. 269–284], under the heading “Observatio de aequatione sexti gradus ad quam aequationes sexti gradus revocari possunt,” and the second, that of Kronecker in the note “Sur la résolution de l’équation du cinquième degré,” *Comptes Rendus*, t. XLVI. (1858), pp. 1150–1152. Jacobi’s equation is closely connected with that obtained by Malfatti in 1771, see Brioschi’s paper “Sulla risolvente di Malfatti per l’equazione del quinto grado,” *Mem. R. Ist. Lomb.*, t. IX. (1863); but the characteristic property first presents itself in Jacobi’s form, and I think the equation is properly described as Jacobi’s resolvent equation. The other equation has been always known as Kronecker’s resolvent equation; it belongs to the class of equations for the multiplier of an elliptic function considered by Jacobi in the paper “Suite des notices sur les fonctions elliptiques,” *Crelle*, t. III. (1828), pp. 303–310, see p. 308, [*Ges. Werke*, t. I., pp. 255–263, see p. 261]: say Kronecker’s equation belongs to the class of Jacobi’s Multiplier Equations. We have in regard to it the paper by Brioschi, “Sul metodo di Kronecker per la risoluzione delle equazioni di quinto grado,” *Atti Ist. Lomb.*, t. I. (1858), pp. 275–282, and see also the “Appendice terza” to his translation of my *Elliptic Functions* (Milan, 1880): it seems to me however that the theory of Kronecker’s equation has not hitherto been exhibited in the clearest form.

I consider the forms

12345	13524
13254	12435
24315	23541
35421	34152
41532	45213
52143	51324

which, if in the first instance the figures are regarded as points, represent the twelve pentagons which can be formed with the points 1, 2, 3, 4, 5; each form in the right-hand column is derived from the corresponding form in the left-hand column by stellation, say we have $13524 = S\,12345$, and so in other cases.

A pentagon is in general reversible, but we sometimes consider it as irreversible (viz. we distinguish between the pentagons 12345 and 15432); when this is so, we write $15432 = R\,12345$, and we have thus twelve new forms, in all twenty-four forms. The symbols R , S are such that $R^2 = 1$, $S^2 = R$, $RS = SR$, $S^4 = 1$. But for a reversible pentagon, there is no occasion to use the symbol R , and we have simply $S^2 = 1$.

In a somewhat different point of view, we may for an irreversible pentagon write

12345	to denote any one of the forms	12345, 23451, 34512, 45123, 51234,
$R\,12345$	to denote any one of the forms	15432, 54321, 43215, 32154, 21543,

and for a reversible pentagon 12345 to denote any one of these same ten forms.

Each pentagon gives thus ten forms, viz. we have in all 120 forms which all are the different arrangements of the five figures. But further regarding the arrangement 12345 as positive, then the forms in the left-hand column are each of them positive, and the ten forms derived from any one of these are each of them positive, that is, the forms in the left-hand column give all the 60 positive arrangements of the five figures: and similarly the forms in the right-hand column give all the 60 negative arrangements of the five figures.

Taking 1, 2, 3, 4, 5 to denote any quantities, or say the five roots x_1, x_2, x_3, x_4, x_5 of a quintic equation, we regard 12345, . . . , as denoting functions of these roots: in particular, 12345 may denote a cyclic reversible function, the analogue of the reversible pentagon, or it may denote a cyclic irreversible function, the analogue of the irreversible pentagon.

Jacobi's resolvent equation.

The most simple course is to take 12345 a cyclic reversible function of the roots x ; a root of the resolvent equation is then $12345 - 13524, = (1 - S) 12345$, and the six roots are

$$z_1 = (1 - S) 12345,$$

$$z_2 = (1 - S) 13254,$$

$$z_3 = (1 - S) 24315,$$

$$z_4 = (1 - S) 35421,$$

$$z_5 = (1 - S) 41532,$$

$$z_6 = (1 - S) 52143.$$

Here effecting on the roots x any positive substitution whatever, we permute *inter se* the roots z ; but effecting on the roots x any negative substitution whatever, then [the discussion continues on the following page, outside the present block].